

PROFILE Research engineer specialising in generative models, computer vision and large-scale ML systems.

INDUSTRIAL
EXPERIENCE

Meta/Facebook, London

Tech lead, Applied AI

May 2026 to present

- Leading a team of 8 scientists/engineers enhancing the reasoning capabilities of large-scale models.

Research engineer, XR

November 2024 to May 2026

- Contributed to [3D AssetGen](#) and [WorldGen](#) projects, training 3D super-resolution and mesh-topology models and designing workflows that improve geometric fidelity and usability of generated worlds and assets.
- Owned key parts of synthetic 3D data acquisition and processing (vendor coordination, specifications, validation and pre-processing pipelines, fast data loading) at PB scale; collaborated with Horizon teams on requirements for creator workflows and integration with Horizon experiences.
- strengthened team engineering practices (data frameworks, GPU utilisation, on-call playbooks), enabling reliable scaling of generative-model training and inference.

Research engineer, FAIR / GenAI

July 2019 to November 2024

- Co-authored 13 publications at CVPR, ICCV, NeurIPS and ECCV on 3D reconstruction and generation models, advancing fundamental research in 3D vision and neural rendering.
- Co-led [uCO3D](#), combining large-scale 3D video collection with training state-of-the-art text-to-3D generative models.
- Designed and implemented [Implicitron](#) and contributed core components to [PyTorch3D](#), resulting in a modular open-source new-view-synthesis framework and tooling widely used internally and externally.

ML engineer, Ads ranking

May 2017 to July 2019

- Improved ads ranking pipelines and models for targeting objectives such as longer attribution windows and incremental conversions, using experimentation and causal analysis to increase the relevance of ads delivery.

BlippAR, London

Computer vision research engineer

March 2016 to May 2017

- Optimised mobile AR tracker (C++) and built (in a team of two) an urban positioning and mapping backend for the AR City app.

Artec Group, Moscow

Research and development engineer

September 2014 to March 2016

- Improved the accuracy of 3D-based face verification by fusing depth with images and training models for representation and metric learning (Theano, C++).

MSU Graphics & Media Laboratory, Moscow, Russia

Researcher / C++ developer

August 2008 to June 2011

- Research in object detection and segmentation of 2D and 3D geospatial data; SDK integrated in a road surface QA system.

NetCracker, Moscow, Russia

Part-time Java developer

July 2007 to August 2008

- Maintained a telecom infrastructure visualisation tool and integrated it with back-end operations support systems.

PROFESSIONAL
SKILLS

Tech stack: Python; PyTorch; NumPy; pandas; Blender; scalable data processing pipelines.

CV / graphics: image-to-3D; diffusion models; CNNs; SfM / SLAM; novel-view synthesis; 3D representations; object detection and segmentation in 2D and 3D.

Machine learning: classical ML; probabilistic graphical models; Bayesian optimisation; causal inference; recommender systems; transformers.

EDUCATION

Lomonosov Moscow State University

PhD in physics and mathematics, ML focus

October 2010 to March 2015

- Thesis: *Structural learning methods for collective labelling problems* (computer vision, structured prediction, weak supervision), advisor: [Dr. Dmitry Vetrov](#)
- Google Summer of Code internship: object detection and local descriptors in Point Cloud Library (PCL).
- Teaching assistance for computer vision and ML courses: authored and graded assignments for classes of 250 students.

specialist in applied math and computer science **September 2005 to June 2010**

- GPA: 5.0 out of 5.0; thesis: *Automated object detection in laser-scanning data*

ACADEMIC
SERVICE

Reviewer: CVPR, ICCV, ECCV, ACCV, NeurIPS (1–3 times a year since 2015).

SELECTED
PUBLICATIONS

- D. Novotny*, **R. Shapovalov***, A. Vedaldi. Canonical 3D Deformer Maps: Unifying parametric and non-parametric methods for dense weakly-supervised category reconstruction. *NeurIPS* 2020.
- J. Reizenstein, **R. Shapovalov**, P. Henzler, L. Sbordone, P. Labatut, D. Novotny. Common Objects in 3D: Large-scale Learning and Evaluation of Real-life 3D Category Reconstruction. *ICCV* 2021, oral presentation, **Best Paper Honourable Mention**.
- R. Shapovalov**, D. Novotny, B. Graham, P. Labatut, A. Vedaldi. DensePose 3D: Lifting Canonical Surface Maps of Articulated Objects to the Third Dimension. *ICCV* 2021, **Oral Presentation**.
- P. Henzler, J. Reizenstein, P. Labatut, **R. Shapovalov**, T. Ritschel, A. Vedaldi, D. Novotny. Unsupervised Learning of 3D Object Categories from Videos in the Wild. *CVPR* 2021.
- R. Shapovalov**, Y. Kleiman, I. Rocco, D. Novotny, A. Vedaldi, C. Chen, F. Kokkinos, B. Graham, N. Neverova. Replay: Multi-modal Multi-view Acted Videos for Casual Holography. *ICCV* 2023.
- Y. Siddiqui, T. Monnier*, F. Kokkinos*, M. Kariya, Y. Kleiman, E. Garreau, O. Gafni, N. Neverova, A. Vedaldi, **R. Shapovalov***, D. Novotny*. Meta 3D AssetGen: Text-to-Mesh Generation with High-Quality Geometry, Texture, and PBR Materials. *NeurIPS* 2024.
- M. Chen, **R. Shapovalov**, I. Laina, T. Monnier, J. Wang, D. Novotny, A. Vedaldi. PartGen: Part-level 3D Generation and Reconstruction with Multi-View Diffusion Models. *CVPR* 2025.
- X. Liu, P. Tayal, J. Wang, J. Zarzar, T. Monnier, K. Tertikas, J. Duan, A. Toisoul, J. Zhang, N. Neverova, A. Vedaldi, **R. Shapovalov***, D. Novotny*. Uncommon Objects in 3D. *ICCV* 2025.
- W. Li, A. Toisoul, T. Monnier, **R. Shapovalov**, R. Ranjan, P. Tan, A. Vedaldi. MeshFlow: Efficient Artistic Mesh Generation via MeshVAE and Flow-based Diffusion Transformer. *CVPR*. 2026. **Highlight**.
- D. Wang, X. Xiang, K. Sohn, T. Monnier, Y.-Y. Yeh, T. Nguyen-Phuoc, J. Zhang, Y. Fan, A. Toisoul, H. Jung, P. Dhar, M. Bunnell, N. Sarafianos, C. Zou, **R. Shapovalov**, A. Vedaldi, R. Ranjan. AssetGen: Deployable 3D Asset Generation at Interactive Speed. *arXiv preprint*. 2026.